Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across demographics, product preferences, and subscription behavior.

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Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Values

Review Rating column

$59.8

Avg. Purchase

Key features: Age, Gender, Location, Subscription, Category, Purchase Amount, Discount, Shipping, Review Rating.

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Data Preparation in Python

Load Data

Import via pandas

Clean Data

Median imputation for Review Rating

Engineer Features

Create age_group and purchase_frequency_days

Load to PostgreSQL

Write cleaned DataFrame to DB

Columns standardized to snake_case; redundant promo_code_used dropped; cleaned DataFrame loaded into PostgreSQL.

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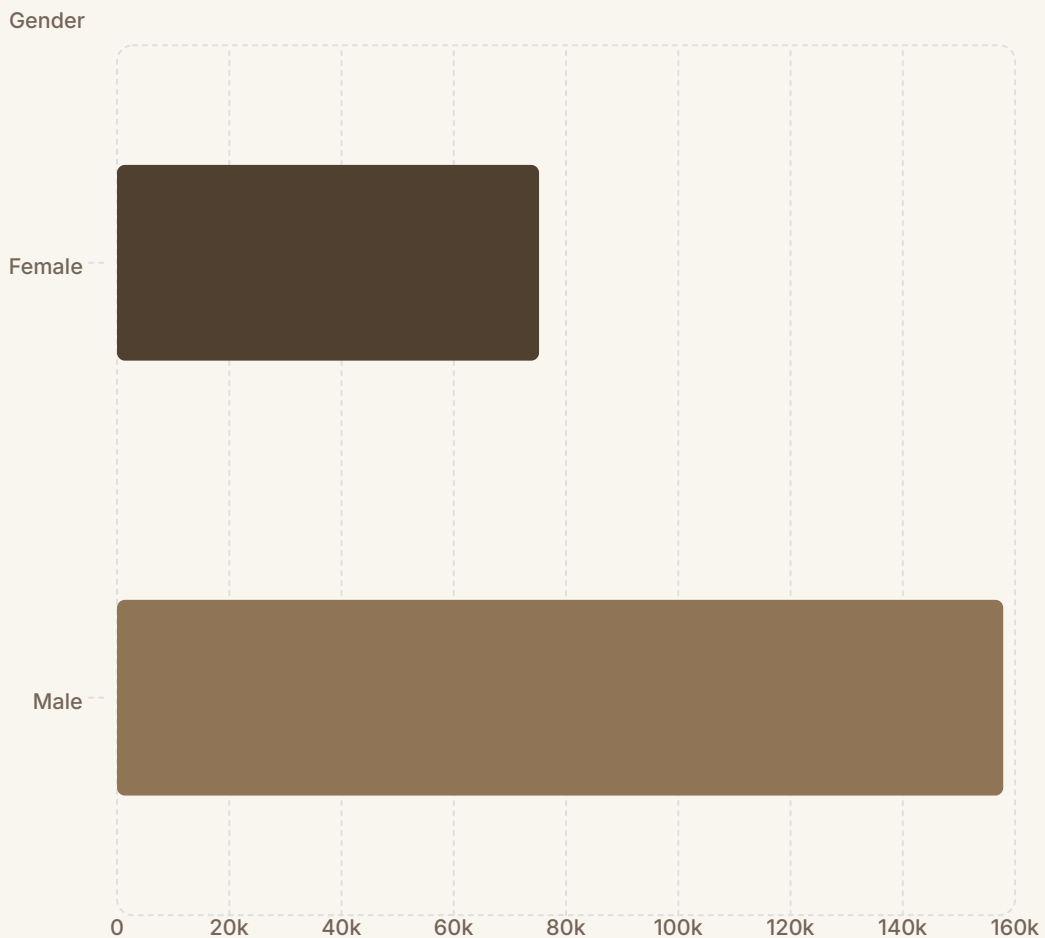
Revenue by Gender



Key Insight

Male customers generated **2.1× more revenue** than female customers — a significant gap worth investigating.

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Product & Shipping Insights

Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Shipping Type Comparison

Type	Avg Spend
Standard	$58.46
Express	$60.48

Express shipping users spend ~$2 more on average.

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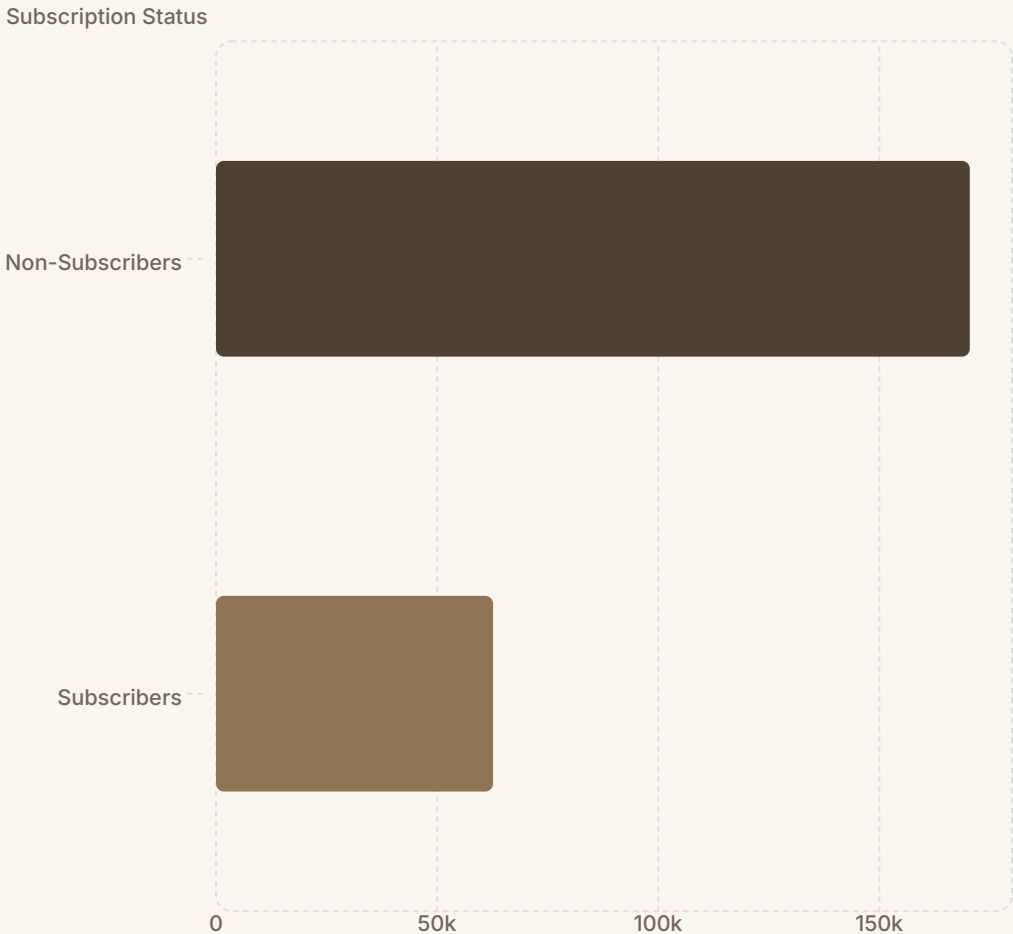
Subscription Behavior



Subscriber Profile

27% subscribed (1,053 customers)

Avg spend nearly identical: **$59.49** vs **$59.87**

Non-subscribers drive **73%** of total revenue

839 customers used discounts yet spent above average — high-value deal-seekers.

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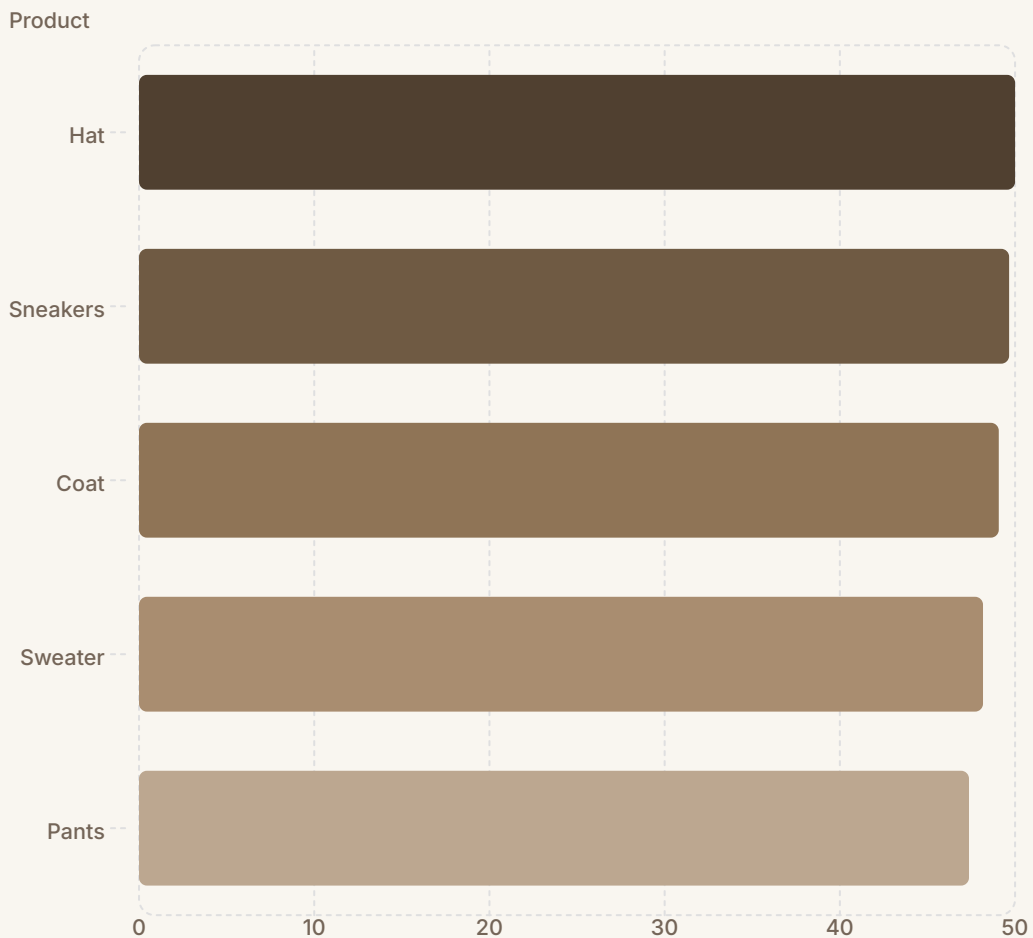
Discount-Dependent Products



Key Observation

All top 5 discount-heavy products have **~47–50%** of purchases made with a discount applied.

Overall, **43%** of all purchases used a discount.

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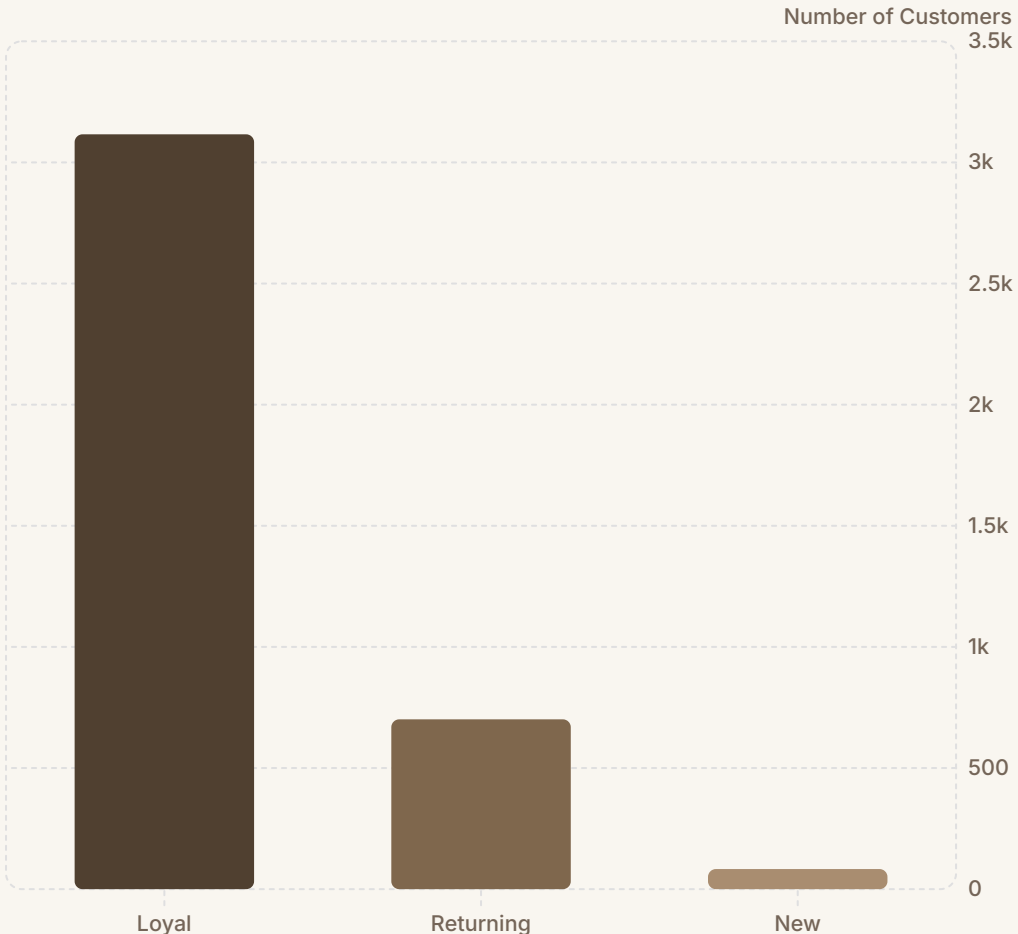
Customer Segmentation



Segment Breakdown

Loyal: 3,116 customers — dominant segment

Returning: 701 customers

New: Only 83 customers — acquisition opportunity

958 repeat buyers (5+ purchases) are subscribers — strong loyalty-subscription link.

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Power BI Dashboard

KPIs

3.9K customers · $59.8 avg purchase · 3.75 avg rating

Revenue by Category

Clothing $104K · Accessories $74K · Footwear $36K · Outerwear $19K

Filters

Gender, Subscription, Category, Shipping Type, Age Group

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow Loyal segment

Review Discount Policy

Balance sales boost with margin control

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

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